Wanted for: Indecision, General Wishy-Washiness, Trouble Telling *up* from *down*

Description: There is some information about the composite system, and some about each subsystem. Incomplete in each case.

State-Vector: $\sqrt{0.6}|ud\rangle - \sqrt{0.4}|du\rangle$

Normalization: $\psi^*_{uu}\psi_{uu} + \psi^*_{ud}\psi_{ud} + \psi^*_{du}\psi_{du} + \psi^*_{dd}\psi_{dd} = 1$

Density Matrix:
Full Composite System: $\rho^2 \neq \rho$, and $Tr(\rho^2) < 1$.
Alice's Subsystem: $\rho^2 \neq \rho$, and $Tr(\rho^2) < 1$.

Wave Function: Not Factorized: $\psi(a, b)$

Expectation Values:

$$\langle\sigma_z\rangle = 0.2$$
$$\langle\sigma_x\rangle, \langle\sigma_y\rangle = 0; \langle\tau_z\rangle = -0.2$$
$$\langle\tau_x\rangle, \langle\tau_y\rangle = 0$$
$$\langle\tau_z\sigma_z\rangle = -1$$
$$\langle\tau_x\sigma_x\rangle = -2\sqrt{0.24}$$

Correlation: $\langle\sigma_z\tau_z\rangle - \langle\sigma_z\rangle\langle\tau_z\rangle = -0.96$ for this example. For partially entangled states in general, correlation is between −1 and +1, but not exactly 0.

Exercise 7.11: Calculate Alice's density matrix for σ_z for the "near-singlet" state.